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Waste Separation Learning Application Based on Octalysis Framework

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Abstract

In answering the ever-growing trash problem and other problem stemming from it especially in Indonesia, an educational game is made to educate people on how to do waste separation, a concept that is not known by many but have a big impact in the step of recycling overall. “EcoHero” mainly taught about waste separation as a way to help the process of recycle, in hopes of reducing overall waste produced. Player will have to collect many types of trash scattered on the ground then separate it based on the waste type, organic, non-organic, and hazardous, to accumulate points and pass the stage. EcoHero is made for mobile device on Unity by utilizing Game Development Life Cycle as the development method and Octalysis framework as the baseline for features. Features and game experiences were tested on 34 participants with age ranging from 18 to 25 years old. Data were collected using forms and results returned good, with application evaluation scoring 85%, in which the game is working properly as intended and game experience scores ranging between 75% to 91%, in which the game gives good experience when played.

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Keywords: waste separation; serious game; Game Development Life Cycle; Octalysis framework; Game Experience Questionnaire;

1. Introduction

Indonesians produce millions of tons of garbage every single day. In fact, one of the largest landfills in terms of annual volume of waste is located in Indonesia, specifically Bantargebang in Jakarta [1]. According to Plastic Bank

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Indonesia in 2022, 4.9 million tons of plastic waste out of 7.8 million tons are not handled properly [2]. The massive amount of trash also reflects with some of the problems happening in Indonesia, with the most apparent problem being flash flood in many cities, mainly caused by household wastes blocking the sewage [3], [4].

One of the known methods for waste management is by recycling, which Indonesian governments have tried raising awareness about [5]. Not all the waste can be recycled; therefore, waste separation is an essential step to fasten the process of recycling [6]. Optimally, the trash should already be separated within each household. However, most of Indonesian residents do not conduct waste separation and instead, this step is done at temporary dumping ground and landfills, making it hard for the scavengers to retrieve recyclables [7].

Video games are very popular among people, from young'uns to old folks. In fact, video game industry generated approximately \$455 billion in 2024 [8]. Aside from entertainment purposes, video games can be an effective approach for learning many things. For example, a research projects studied about using video games as a method to improve decision making, cognitive skills, and time management [9]. Serious game is the use of games to fill needs other than pure entertainment, which is perfect for making learning experience more exciting with the entertaining game aspects [10].

With the rising trends of video games, one would bound to think about how are video games created. Video games are like a craft, with combination of art, music, programming, acting, and management, integrated into one software that we all know as video game. However, these things cannot be mixed on a whim, they have to be prepared and inserted carefully. Hence, exists a baseline for creating game, known as Game Development Life Cycle (or GDLC for short). GDLC helps game developer by organizing phases during development to make sure the process is arranged properly and optimally, to see what and why changes are needed on certain parts of the game based on feedbacks and tests [11].

Creating an interesting a game is about manipulating one's motivations and fulfilling them with features in the game. Those motivations can be called as Core Drives, and it exists within every individual. To understand these Core Drives better, exists Octalysis framework which is a gamification framework that grouped many game elements divided into eight Core Drives with each characteristic. Those Core Drives are Epic Meaning & Calling, Development & Accomplishment, Empowerment of Creativity & Feedback, Ownership & Possession, Social Influence & Relatedness, Scarcity & Impatience, Unpredictability & Curiosity, and Loss & Avoidance. Octalysis framework provides in-depth analysis of each Core Drive representation within the game [12].

Octalysis framework have seen many uses within the topic of video game, ranging from game reviews to game creations. One of the famous examples and a big influence to this framework is Diablo II, bringing Core Drives like Epic Meaning & Calling, represented by Humanity Hero in which the player will go on an epic journey against the demons and save the world. Another Core Drives brought by Diablo II comes in the form of Creativity & Feedback, with Plant Picker as its representative, allowing players to build their character with desired class like Hunters or Berserker [13].

There's also been many instances of Octalysis framework being implemented as the baseline of game creation and gamification. For examples, [14] used Octalysis framework as the basis for game-based Japanese language learning. Another research also implemented Octalysis framework in creating a mobile learning application for English [15]. Octalysis framework is also implemented in the design of time management application [16]. Researches mentioned above used Octalysis framework to create an interesting gamified applications by utilizing Core Drives that brings up the motivation of people.

The aim of this research is to create a game that advertise waste separation to a broader audience, specifically Indonesian, while implementing Octalysis framework paired with Game Development Life Cycle as the development method to spark interest on the intended audience target. Garbage problem in Indonesia have grown into a worrying level. Hopefully, this research will help to alleviate the problem by building awareness of the importance of waste separation.

2. Related Works

The lack of waste separation knowledge in Indonesia has been very worrying when paired with the fact that Indonesia is one of the largest garbage producing country. There have been many attempts from people to share the information about waste separation and its benefits, one of the examples being [17] which was done in 2021 in a populated area. Other research also attempted to introduce waste separation in the early days of students [18] for them to build a good habit in sorting waste and doing their part in waste management. Research was also conducted to one of the universities in Indonesia [19] with the hope of helping solid waste management starting from the campus ground.

By using video game as a media for teaching, it helps in keeping the audience engaged, making it much lighter for people to absorb knowledge shared by the game. Using game as a learning method for waste management have proven to be very fruitful, especially towards younger audience. One example is [20] which used a trash tracking game to teach about waste separation as early as kindergarten. Another example is [21] which made a game to separate waste by its type and tested to elementary school students.

3. Methods

3.1. Game Development Life Cycle

Game Development Life Cycle (GDLC) acts as a guideline for game development process. GDLC consists of six development phases, which are Initiation, Pre-Production, Production, Testing, Beta, and Release, as shown in Fig. 1. [11].

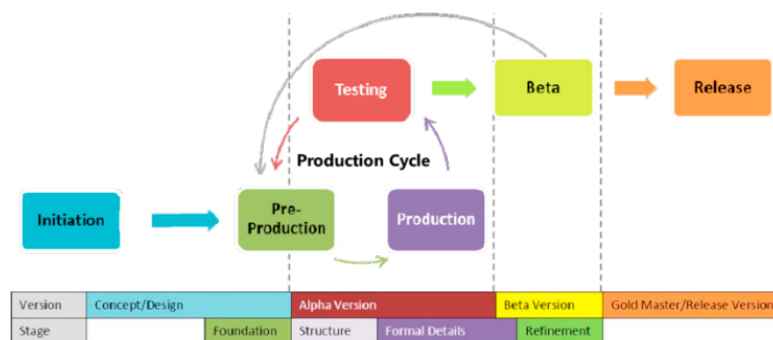


Fig. 1. Game Development Life Cycle

The first phase in GDLC is Initiation, in which the rough concept of the game is discussed. The output of this phase is the game concept and simple descriptions of the game.

Pre-Production is the next phase after Initiation, and also marks the start of Production Cycle in GDLC. This phase involves the creation of game design and game prototype. Creating game design focuses on defining game genre, general gameplay, game mechanics and its features, which will then be compiled to game design document (GDD). After that, the prototype is made to assess the game design and ideas. Prototype will focus on the foundations and structure on the first iteration, while next iterations will focus on the details and refinement. Development will continue to next phase after all the revisions been approved and documented in GDD.

Production is the core process in game creation, revolving around the more technical steps like assets creation, source code creation, and the integration of both. Production phase also includes game balancing in terms of difficulty and performance of the game. The result of this phase will then be tested by selected people in the next phase.

After completing the production phase, testing phase comes next. Testing in this context means the conduct of internal testing for the game where the selected group of people will test the game and give out feedbacks on their experience while playing the game. Most common thing to look out for in the testing phase are bugs, loopholes, dead-ends, difficulties and accessibility aspects. The results of this phase will determine whether the prototype can advance to the next phase or whether the reiteration of production cycle is needed.

Beta phase is similar to testing phase, but with third-party or external tester. There are two tester selection methods, closed beta and open beta. Closed beta means that the testers will be selected from the participant, while open beta means that anyone who registered can become the participant. The output of this phase are bug reports and user's feedback. If the results are deemed satisfactory, the game will be released, else it will go back to the production cycle to refine the game based on the reports and feedbacks.

Final phase of the game development life cycle is Release phase, where the game will be launched to public. Release phase also includes further steps like maintenance and game expansion in the future.

This research uses the Game Development Life Cycle (GDLC) method for the game creation. The first phase in GDLC is initiation, where discussion about the rough concept of the game is done. Next phase is pre-production which involves the discussion of game designs to fulfill the functions discussed in initiation phase and creation of game prototypes. Production phase is up next, where assets and source codes are created.

The fourth phase is Beta testing, the phase for assessing game's functionality and user experience through feedback from external party. Their perspectives and opinions will be compiled for the developers to improve the game upon. The game will then be released on the itch.io website, available to public in the hope to reach more people.

3.2. Octalysis Framework

There are many famous video game titles that are loved by people or the community. Those video games have many ways to capture players heart despite having differences with each other, motivating new players to start playing and old players to keep playing the game. The creation of Octalysis framework intends to categorize these motivations that can be called as Core Drives.

In Octalysis framework, these Core Drives are divided into eight segments pictured as an octagon. Those core drives are Epic Meaning & Calling, Development & Accomplishment, Empowerment of Creativity & Feedback, Ownership & Possession, Social Influence & Relatedness, Scarcity & Impatience, Unpredictability & Curiosity, and Loss & Avoidance. Each core drive consists of multiple game elements as shown in Fig. 2. [12].



Fig. 2. Octalysis Framework and its Game Elements

First Core Drive is Epic Meaning & Calling, which includes many game elements that can be found at the start of the game. This core drive can be seen as the introduction of the game for the players, story-wise like *Humanity Hero* and *Destiny Child* where the player will take an important role in the lore of that world, or gameplay-wise like *Free Lunch* where new players get many benefits.

Second Core Drive is Development & Accomplishment, which includes game elements about progression step which coincides with human's desire to evolve and achieve. This core drive includes game techniques like Badges and Leaderboard to measure completions and achievements.

Third Core Drive is Empowerment of Creativity & Feedback. This core drive focuses on creativity as a motivational drive to play the game, with game elements such as *Plant Picker* and *Poison Picker* which allow players to experiment with combinations of equipment and weapons, each offering their advantages and disadvantages.

Fourth Core Drive is Ownership & Possession, with game elements triggered by human's desire to collect things and upgrades. Examples of game elements in this core drive are *Virtual Goods* and *Avatar*, collecting rare items and appearances to suit the player's taste.

Fifth Core Drive is Social Influence and Relatedness. This core drive utilizes human's nature as a social being with game elements like *Group Quest* for players to complete missions together and *Mentorship* for experienced players to help new players in learning the game.

Sixth Core Drive is Scarcity & Impatience, pressing on the urgency of matters as a motivation for players to complete an action. Examples of game elements included in this core drive are *Prize Pacing* where a player would need to perform an action for extra rewards and *Count Down Timer* where there's a limited amount of time for player to earn certain items or receive certain rewards.

Seventh Core Drive is Unpredictability & Curiosity, using human's nature of curiosity and uncertainty to whether give a punishment or special prize. *Rolling Rewards* is one of the examples of game element for this core drive, allowing players to sacrifice something in order to get a varying degree of rewards from commons to rare items.

Eighth and final Core Drive is Loss & Avoidance. This core drive motivates player through the fear of something going wrong. Game element example of this core drive is *Progress Loss* which dissuades players from making mistakes by taking away progress after accidents.

GDLC is one of the most common methods used to help in creating a game, because it flows perfectly between the creation phase and testing phase. Octalysis framework is known for its usage in creating a gamification-based system to help create an interesting game through the usage of game elements as motivation to play. This [22] research utilized GDLC for the development of the game while applying the Octalysis framework to analyze the implemented features in the pre-production stage of the development.

3.3. Evaluation Method

We used a dependent group approach for this experiment and convenience sampling to choose the participant. There are 34 participants from the open beta, which mainly consists of college students from Jakarta, Indonesia, with age ranging from 18 to 25 years old. Participants will play our game for 15 minutes before filling in the application evaluation form and game experience questionnaire.

Usability test will be conducted with 10 questions regarding application evaluation with Likert 5-point scales of agreement, ranging from 1 to 5 with 1 as "strongly disagree" and 5 as "strongly agree" as the scoring metric. Game experience questionnaire will also be handed with 25 questions representing 7 components in GEQ Core Module, with

scoring metric ranging from 1 to 4 with 1 as “not at all” and 4 as “extremely”, referring to experience had during testing. To ensure the accuracy and consistency of the questionnaire, we will do validity and reliability tests.

4. Results

4.1. Development Result

Following the Game Development Life Cycle, the development starts with the Initiation phase, where discussion was done with all the developers. The game is titled “EcoHero”, in which players will play as a character with a goal to collect the trash scattered on the field and properly separates them into three types of trash, which are organic, non-organic, and hazardous. The goal is to achieve the target score within the time limit.

Next up is the Pre-Production phase. The prototype was made to fulfill the requirements of functions discussed in the initiation phase and game designs discussed in this phase. Added some features like Almanac and packed trash to help players in understanding the game further and add diversity in gameplay. Almanac contains all the trash that can be found in the game, where players can check the type of waste and see whether they have correctly separated the waste, shown by the highlights. Packed trash is a gimmick where some of the trash contains multiple trash after it has been unpacked, comes in a form of boxes and plastic bag.

In production phase, assets and source codes are created. Assets made includes the character, user interface, main menu, buttons, the field, almanac, and all kinds of trash.

There are some very simple buttons in the main menu which can be seen below in Fig. 3. “Start” button will lead to level selection screen, there are 3 available levels on release with the main difference being the amount of trash varieties. “Settings” button will open the configuration for volumes, and “Quit” button will quit the application when pressed.



Fig. 3. Main Menu Screen

Fig. 4(a). below shows the game scene. Main objective on this stage is to pick up trash that spawns across the map. Analog on the left to move the character, hand button on the right to collect the trash on the floor when the character is on top of it. Hand button can also be used to interact with the table to start the waste separation process. There are 2 numbers on the screen, top left shows your score which you gain for every waste separation process done. The top middle one refers to the timer before the stage ends. Pause button on the top right will pause the game, with players given choice to resume, change settings, or go back to main menu.

Waste separation screen can be triggered by pressing the hand / action button near the table. Can be seen below in Fig. 4(b)., waste separation scene consists of the 3 colored trash bin on the right and almanac on the bottom right. Players can drag the trash from the inventory freely on the table or straight to the correct trash bin. For every waste separated correctly, player will get +100 points, and -50 for every incorrect separation.

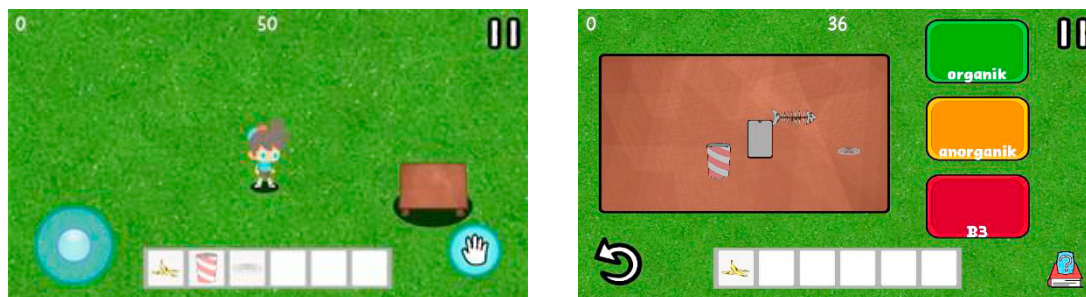


Fig. 4. (a) Game Scene; (b) Waste Separation Scene

Almanac can be accessed from the waste separation table at the bottom right of the screen with book and a question mark symbol. This almanac will show the type of each individual trash and its name. Trash that has not been found will be darker, while the brighter one means that player have put the trash in the trash bin. Almanac can be seen in Fig. 5(a). below.

There are special kinds of trash, namely packed trash. Packed trash is a compacted trash that consists of multiple trash, often the combination of different types of trash. Players must drag the packed trash from the inventory to the table, tap on the trash and press the unpack button. Fig. 5(b). below shows how to unpack the packed trash.

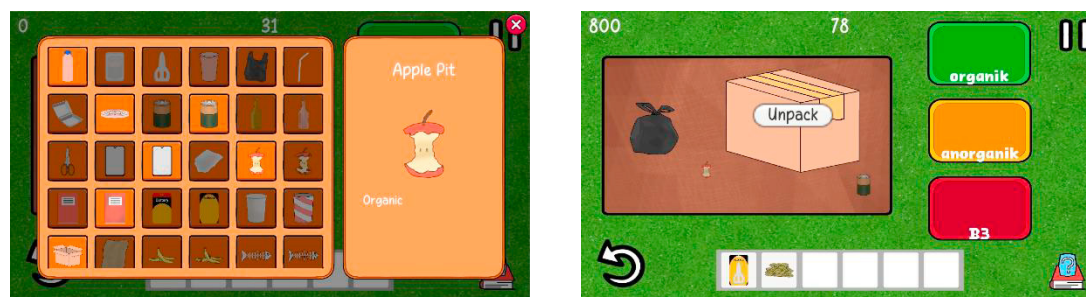


Fig. 5. (a) Almanac; (b) Packed Trash

The game element used to represent first core drive is Higher Meaning, in which players are tasked to help with waste separation in order to make the field cleaner. For the second core drive, Points are portrayed by the score you get for a successful attempt of separating wastes. Real-Time Control is the game element used in this game by players always in control of the character when collecting and separating wastes. Game element in EcoHero that fits the fourth core drive well is Collection Set, where players can collect all types of garbage available and check them in almanac.

Fifth core drive which is Social Influence and Relatedness is sadly not implemented in EcoHero due to the nature of the game being a singleplayer game.

Sixth core drive is represented by Count Down game element, with each stages having their own time limit before the stage ends. Mini Quests represents the seventh core drive in EcoHero, because of how boxes work as a container for more garbage, technically having bigger chance to net more points with many kinds of garbage. For the final core drive, Progress Loss is the game element used, with score penalty for every mistake in separating the wastes.

4.2. Evaluation Result

Usability test questionnaire is handed to respondents with questions related to “EcoHero” features and performance, with questions and results can be seen in Table 1 below. Using Likert Scale 1-5, the scoring metric follows as such:

each input will be multiplied with their respective scale, with the maximum score of perfect score (in this case, 5) multiplied by the total of 34 respondents, resulting in 1700 as 100%.

Table 1. Usability Test Questionnaire and Results

Number	Questions	Scoring					Total Score
		1	2	3	4	5	
1	This game helps in learning knowledge about waste separation.	1	0	0	10	23	156
2	This game is entertaining and exciting	0	1	3	16	14	145
3	The buttons in this game have consistent shapes and colors	0	0	0	15	19	155
4	Always feeling in control of the game because the buttons are responsive	0	3	3	15	13	140
5	Clear information available for further action after an action has been performed	0	2	3	12	17	146
6	Instructions of how to play the game are clear and easy to understand	1	3	7	10	13	133
7	The game can run well without any obstacles such as bug	1	0	7	12	14	140
8	Design and appearance of the game is visually appealing	0	0	8	12	14	142
9	Clear feedback after doing something in the game	1	0	6	14	13	140
10	Actions such as restarting levels and exiting the game can be done easily	0	0	4	12	18	150
Questionnaire Results		(1447 / 1700) * 100% = 85%					

Validity and Reliability test is also done to ensure the variables used are valid and reliable. Validity test is done using the item - total correlation and results returned that all the variables are valid $r > 0.344$. Cronbach's alpha test is used for the reliability test and results confirmed that the ten variables have good internal consistency overall with $\alpha = 0.767$. Results of the validity and reliability test can be seen in Table 2 below.

Table 2. Validity Test using Item - Total Correlation and Reliability Test using Cronbach's Alpha Test

Question Number	r_{xy}	r_{table} (5% significance)	Validity	Alpha	Correlation (r) with other items	Alpha without item
Q1	0.43	0.344	Valid	0.767	0.297	0.764
Q2	0.65	0.344	Valid		0.511	0.738
Q3	0.49	0.344	Valid		0.422	0.754
Q4	0.59	0.344	Valid		0.456	0.744
Q5	0.63	0.344	Valid		0.514	0.736
Q6	0.8	0.344	Valid		0.696	0.702
Q7	0.63	0.344	Valid		0.499	0.738
Q8	0.56	0.344	Valid		0.402	0.751
Q9	0.42	0.344	Valid		0.269	0.771
Q10	0.39	0.344	Valid		0.275	0.766

The Game Experience Questionnaire (GEQ) [23] was also used to assess player's experience after playing the game. The questionnaire consists of 25 questions with 7 components, each question's score ranging from 1-4, with 1 being disagree and 4 being agree. Results can be seen in Table 3 below.

Table 3. Game Experience Questionnaire Results

No.	Components	Average	Percentage
1	Competence	3.66	91.66%
2	Sensory and Imaginative Immersion	3.1	77.5%
3	Flow	3	75%
4	Tension / Annoyance	2	50%
5	Challenge	3.125	78.125%
6	Negative Affect	2.25	56.25%
7	Positive Affect	3.5	87.5%

Game experience questionnaire can be divided by positive and negative aspect. Positive aspect includes Competence, Sensory and Imaginative Immersion, Flow, Challenge, and Positive Affect. Competence refers to player's feeling competent in clearing the game. Sensory and Imaginative Immersion means that the game is pleasing to human senses, such as visual, story and sound. Flow in this context refers to how the game pulls your focus more towards the game. Challenge refers to the difficulty level of challenges that appears in the game like puzzles and problem solving, with higher percentage representing the balance in challenges difficulty. Negative aspect includes Tension / Annoyance and Negative Affect. Tension or annoyance refers to things in the game that are not fun and annoying to deal with. Positive and Negative Affect refers to how the game effects player's mood and feeling. The desired result of using Game Experience Questionnaire is to have higher percentage on positive aspects and lower percentage on negative aspects.

5. Conclusion

The development of EcoHero, a waste separation learning application, as one of the ways to help the process of recycle as a solution to the ever-growing problem of trash problem in Indonesia. Prototype is built with utilizing Octalysis framework as the base for design and GDLC as the development method. Testing of the game returns with good results in the technical aspect like game performance and user experience. Quality assessment scored 85% and game experience questionnaire returned good results ranging from 75% to 91% on the good aspects.

This research is clearly far from perfect, with many limitations and challenges found during the research. Things like performance issue on some lower end devices, small amount of application testing sample, and the lack of content within the game. This research can be improved upon in the future, with more variations of contents, multiplayer mode, better application performance, and wider reach for testing. Moreover, for the future research, we would like to see more game-based learning implemented in more fields, as well as integrated into the educational curriculum to benefit more people and to make learning much more enjoyable.

Acknowledgement

The paper only covers the evaluation result, with the raw result will be made available upon request to the authors. Each author presents contributions as follows: Swasa Raditya Tirta focuses on research ideation, data gathering, data analysis, presentation of data analysis, and article writing; Galih Dea Pratama focuses on research supervision and article proofreading.

Data Availability

The authors: Swasa Raditya Tirta and Galih Dea Pratama. We demonstrate our commitment to data openness and transparency. To facilitate further research, we have made the data used in their study publicly available.

The data used by the author can be opened via link below:

- Github Repository
- Downloadable Game
- Questions and Respondent Data

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